# Project Title: Spam Call Detection

A Project Report

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# Abstract

This project presents a machine-learning-based Spam Call Detection system designed to accurately classify incoming calls as spam or legitimate. Spam calls have become a major threat to user privacy, financial security, and communication efficiency. With the rapid growth of automated dialling systems and evolving fraudulent calling techniques, traditional rule-based filtering has become ineffective. The proposed system analyses call metadata, user-reported patterns, number frequency, and text-based information such as transcripts or caller descriptions to improve detection accuracy. Using preprocessing techniques, the dataset is cleaned, normalized, and transformed into numerical features through methods such as TF-IDF and vectorization. Multiple machine learning models, including Logistic Regression, Random Forest, and Support Vector Machine, are trained and evaluated to determine the most efficient classifier. The model performance is measured using accuracy, precision, recall, and F1-score, ensuring balanced and reliable detection of both spam and legitimate calls. The proposed system demonstrates strong potential for integration into telecom services and smartphone applications, offering an efficient and scalable solution to reduce unwanted and harmful spam calls.

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# 1. Introduction

## 1.1 Overview

Spam calls, which are also called robocalls, fraudulent calls, or unwanted marketing calls, have gone up a lot because of improvements in automated calling systems and VoIP technologies. These calls are annoying, cost money, and put your privacy at risk. Traditional rule-based systems can't keep up with new scam tactics, which is why machine learning is such a useful tool for automatically finding spam patterns.

## 1.2 Objectives

- To design a machine learning–based system capable of detecting spam calls.
- To analyze call datasets for extracting meaningful patterns and features.
- To implement classification algorithms to distinguish between spam and genuine calls.
- To improve communication safety by alerting users before answering unknown calls.

## 1.3 Problem Statement

Spam callers consistently modify their tactics to evade traditional detection methods despite the existence of filtering techniques. An intelligent, flexible system that can learn from data and instantly spot new spam patterns is required.

## 2. Literature Review

Due to the sharp rise in robocalls, telemarketing spam, and fraudulent communications, spam call detection has grown in importance as a field of study. Because spam callers regularly switch numbers and adopt new calling patterns, traditional techniques like rule-based filters and blocklists are no longer adequate. In order to create adaptive and intelligent detection models, researchers have consequently concentrated more on machine learning and artificial intelligence techniques. Numerous studies demonstrate how well supervised learning algorithms distinguish between spam and legitimate calls. Because they can handle labelled datasets and extract meaningful decision boundaries, logistic regression, random forest, Support Vector Machine (SVM), and Naive Bayes have been used extensively. Additionally, studies demonstrate that examining call metadata, such as call duration, number frequency, and call time, can greatly increase classification accuracy. Furthermore, research using Natural Language Processing (NLP) of call transcripts shows that text features such as word embeddings and TF-IDF vectors are helpful for identifying fraudulent intent in voice-to-text call data. Deep learning models like Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), which have been used for audio-based spam detection, are examples of recent developments. These models work well against pre-recorded spam messages and robocalls because they extract spectral, tonal, and frequency features from call audio. Additionally, for increased robustness, some researchers suggest hybrid models that combine text, audio, and metadata features.

In comparison to traditional approaches, the literature generally shows that machine learning-based spam call detection offers a scalable, accurate, and adaptive solution. Nevertheless, there are still issues with real-time implementation, dataset availability, and changing spam tactics.

# 3. Methodology

## 3.1 Input Layer

The system begins with collecting four user-level inputs:

- Phone Number
- Call Duration
- Call Frequency
- Message / Voicemail Text

No validation is performed at this stage; raw data is passed directly to the preprocessing module.

## 3.2 Preprocessing Layer

The preprocessing layer transforms raw data into usable features. It consists of three independent processors:

**a)** Phone Number Processor

**b)** Behavioural Processor

**c)** Text Processor

## 3.3 Feature Engineering Layer

All extracted features—

- 6 numerical features, and
- 1000 TF-IDF features

are combined into a 1006-dimensional feature vector.

## 3.4 Normalization Layer

To ensure uniform scaling across heterogeneous features, Standard Scaler performs z-score normalization.

## 3.5 Machine Learning Layer

The core classifier is a Random Forest Model with 100 decision trees.

**Process:**

1. **Bootstrap-sampling**
   Each tree trains on a random subset of the data.

2. **Tree-construction**
   Nodes split using the best feature among a random feature subset.

3. **Ensemble-voting**
   Each tree outputs *spam* or *genuine*.

4. **Majority-vote**
   Final label = class with highest votes.

The model is chosen due to its:

- High accuracy
- Resistance to overfitting
- Ability to handle mixed feature types
- Fast inference time

# 4. Implementation

The implementation of the Spam Call Detection System follows a structured and modular pipeline that transforms raw call information into accurate spam/genuine predictions. The entire system was implemented in Python using libraries such as NumPy, Pandas, Scikit-learn, and NLTK, along with TF-IDF Vectorizer for text processing. This section

## 4.1 Training Pipeline

The training process follows these steps:

1. Load 800-sample dataset
2. Preprocess numeric & textual features
3. Generate 1006-feature vector
4. Encode labels (spam → 1, genuine → 0)
5. Apply Standard Scaler
6. Train-test split (80% training, 20% testing)
7. Train Random Forest (100 trees)
8. Evaluate using accuracy, precision, recall, F1-score
9. Save the final model as .pkl file

This ensures a complete and reproducible machine-learning workflow.

## 4.2 Prediction Pipeline

For new incoming calls:

1. Accept user input
2. Perform identical preprocessing
3. Extract 1006 features

4. Scale using stored scaler
5. Pass through all 100 trees
6. Aggregate votes
7. Compute confidence score
8. Return: Prediction (Spam/Genuine) + Confidence (%)

This enables real-time detection with high accuracy and low latency.

# 5. Results and Discussion

| Algorithm | Accuracy |
|---|---|
| Random Forest | 90% |
| SVM | 85% |
| Logistic Regression | 75% |
| Naïve Bayes | 70% |

### Final Model Evaluation

- Accuracy: 90–95%
- Precision (Spam): 90–95%
- Recall (Spam): 85–92%
- F1-Score: 87–93%

The Random Forest model offers the best performance.

# 6. Conclusion and Future Scope

## 6.1 Conclusion

Using textual and numerical characteristics, the machine learning-based spam detection system effectively and accurately detects spam calls. The model is scalable, reliable, and appropriate for real-time telecom integration.

## 6.2 Future Scope

- Integration with telecom operator APIs
- Deep learning for voice spam detection
- Multilingual message analysis
- Deployment in Android/iOS apps
- City-wide and nation-wide robocall monitoring

# References


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